Batch - 8
Section - 10

A PROJECT TITLE

ON

Travel Time Prediction Using Ridge and Lasso Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

Computer Science and Information Technology

A.Y.: 2026-27

by

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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Name of Title: Travel Time Prediction Using Ridge and Lasso Regression

1. Abstract:

Travel time prediction is an important application in intelligent transportation systems, helping commuters estimate journey duration more accurately by considering multiple factors that influence road travel. This project presents a web-based application, Travel Time Prediction Using Ridge and Lasso Regression, which predicts the expected travel time between a selected starting point and destination.

The application allows users to enter the source and destination locations along with the planned day and departure time. Using mapping services, the system retrieves the route and distance, while weather and traffic information are collected through external APIs. These factors are combined to form the input features for machine learning models.

The project implements two regularized linear regression algorithms, Ridge Regression and Lasso Regression, to predict travel time. These models are trained on historical travel data to learn the relationship between distance, traffic conditions, weather, day of the week, and travel duration. Their performance is evaluated using standard regression metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score to determine the more accurate model.

The developed web application provides users with the predicted travel time, estimated time of arrival (ETA), traffic status, weather conditions, and the recommended route displayed on an interactive map. By integrating machine learning with real-time route and environmental data, the system demonstrates how predictive analytics can improve travel planning and support smarter transportation decisions. This project offers a practical and user-friendly solution while showcasing the effectiveness of Ridge and Lasso Regression in solving real-world travel time prediction problems.

2. INTRODUCTION:

Travel time prediction is an important application of machine learning that helps estimate the duration of a journey more accurately. The prediction depends on various factors such as distance, traffic conditions, weather, day of the week, and departure time. This project develops a web-based application that predicts travel time between a selected source and destination. The system uses route information along with real-time traffic and weather data as input features. Ridge Regression and Lasso Regression models are implemented to generate accurate travel time predictions. The application also displays the estimated arrival time (ETA) and route on an interactive map. This project demonstrates the practical use of machine learning in improving travel planning and transportation efficiency.

3. SOFTWARE REQUIREMENTS:

Operating System: Windows 10/11 or Ubuntu Linux

Programming Language: Python 3.x

Frontend: HTML5, CSS3, JavaScript

Backend Framework: Flask

Machine Learning Library: Scikit-learn

Data Processing: Pandas and NumPy

Visualization: Matplotlib

APIs: OpenRouteService/Google Maps API and OpenWeather API

Development Tool: Visual Studio Code

Web Browser: Google Chrome or Microsoft Edge

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Signature of the Guide

Course Instructor